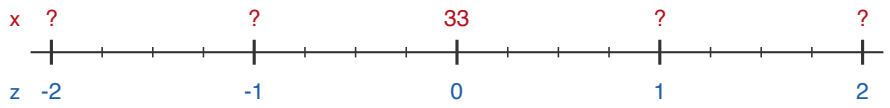
Z-Values

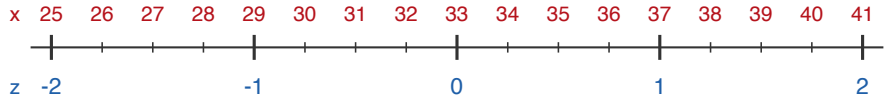
The mean is $\mu = 33$ and the standard deviation is $\sigma = 4$. The z-value of a data value x is $z = \dfrac{x - \mu}{\sigma}$.

a) Fill in the x-values for each z-value on the number line.



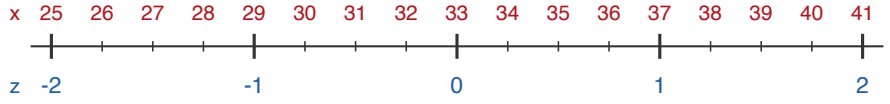
For each data value below, **visually** mark x on the number line and read off its z-value, then **calculate** the z-value with the formula. Check that your two answers match.

b) $x = 39$



Visually: $z =$ _____ Calculate: $z = \dfrac{39 - 33}{4} =$ _____

c) $x = 31$

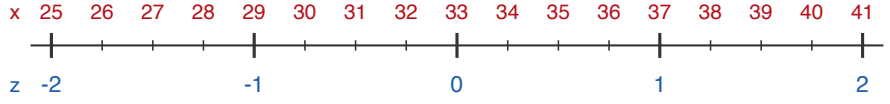


Visually: $z =$ _____ Calculate: $z =$ _____

d) $x = 26$

Visually: $z =$ _____ Calculate: $z =$ _____

e) $x = 36$



Visually: $z =$ _____ Calculate: $z =$ _____